Van der Waerden's theorem

Notes from Carlos' lessons

Source lesson: 2026-04-23 (van der Waerden).

Question on the board. If we colour $1, 2, \ldots, N$ with c colours, can we force a long monochromatic arithmetic progression (AP)? $W(c, L)$ is the least N such that every c-colouring of $\{1, \ldots, N\}$ contains a monochromatic AP of length L.

Theorem 1 (van der Waerden)**.** *$W(c, L)$ exists for all c, L.*

1 $W(2, 3) = 9$

Lower bound ($W(2,3) > 8$). Colour $1, \ldots, 8$ as R R B B R R B B. Red is $\{1, 2, 5, 6\}$ and blue is $\{3, 4, 7, 8\}$. No three elements of $\{1, 2, 5, 6\}$ form an AP (check the four triples $\{1, 2, 5\}, \{1, 2, 6\}$, $\{1, 5, 6\}, \{2, 5, 6\}$: the gaps are $1, 3$ / $1, 4$ / $4, 1$ / $3, 1$, never equal), and $\{3, 4, 7, 8\}$ is the same set shifted by 2. So there is no monochromatic 3-AP. (The board's "Part 1"; verified by brute force.)

Upper bound ($W(2,3) \leq 9$). Every 2-colouring of $1, \ldots, 9$ has a monochromatic 3-AP (the board's "Part 2").

Proof (case analysis). Suppose not. By symmetry of the colours let 5 be red. Each of $\{4, 6\}, \{3, 7\}, \{2, 8\}, \{1, 9\}$ then contains a blue element (otherwise there would be a red 3-AP centred at 5). Consider 4 and 6:

1. *Both blue.* The APs $(2, 4, 6)$ and $(4, 6, 8)$ force 2 and 8 to be red; then $(2, 5, 8)$ is a red AP. Contradiction.
2. *Both red.* Then $(4, 5, 6)$ is red. Contradiction.
3. *Exactly one blue.* The reflection $x \mapsto 10 - x$ preserves APs and fixes 5, so assume 4 is blue and 6 is red.
   - $(5, 6, 7)$: $5, 6$ red, so 7 is blue.
   - $(1, 4, 7)$: $4, 7$ blue, so 1 is red.
   - $(1, 5, 9)$: $1, 5$ red, so 9 is blue.
   - $(7, 8, 9)$: $7, 9$ blue, so 8 is red.
   - $(2, 5, 8)$: $5, 8$ red, so 2 is blue.
   - $(2, 3, 4)$: $2, 4$ blue, so 3 is red.
   - Now $1, 3, 5$ are all red: an AP. Contradiction.

□

2 Why $W(c, L)$ exists

What the board does. It sketches the "columns" idea: cut $\{1, \dots, N\}$ into blocks of length m; a block of length m has one of c^m colourings, so if $N/m \geq W(c^m, L)$, the blocks themselves, viewed as c^m "super-colours", contain an AP of L *identical* blocks ("an arithmetic sequence of identical columns"). For $W(2, 3)$ the board uses blocks of length 5 ($2^5 = 32$ colourings, so any 33 blocks contain two equal ones; the third block of the AP is the next in the progression). It then tries to turn an AP of length L into one of length $L + 1$ and writes a bound involving $W(2, L)$ and $W(2^{W(2,L)}, L)$. I could not read that final inequality reliably, and it is stated for $c = 2$ only, so I give the standard complete argument instead.

Full proof (double induction; the "colour-focusing" argument as in Graham–Rothschild–Spencer; see the sources below). Induct on L: $L = 1, 2$ are trivial. Assume $W(c, L-1)$ exists for *all* c. Fix c.

Definition 2 (Colour-focused APs)**.** APs $A_i = \{a_i + jd_i : 0 \leq j \leq L-2\}$, $i = 1, \dots, r$, are *colour-focused at f* if each A_i is monochromatic, the colours are pairwise distinct, and $a_i + (L-1)d_i = f$ for all i.

Claim 3 (r)**.** *For each $r \leq c$ there is $M(r)$ such that every c-colouring of $[1, M(r)]$ contains a monochromatic L-AP or r colour-focused $(L-1)$-APs with focus $f \leq M(r)$.*

- *Claim(c) finishes the proof.* The focus f has one of the c colours, which equals the colour of some A_i, and then $A_i \cup \{f\}$ is a monochromatic L-AP.
- *Claim(1).* By the induction hypothesis $[1, W(c, L-1)]$ contains a monochromatic $(L-1)$-AP $A_1 = \{a_1 + jd_1\}$ with $a_1 + (L-2)d_1 \leq W(c, L-1)$, hence $d_1 \leq W(c, L-1)$, and its focus $a_1 + (L-1)d_1 \leq 2W(c, L-1) =: M(1)$. One colour-focused AP is just a monochromatic $(L-1)$-AP.
- *Claim($r-1$) $\Rightarrow$ Claim(r).* Let $M = M(r-1)$, $m = M$, and $N = W(c^m, L-1)$. Cut $[1, (2N+1)m]$ into blocks of length m and give each block the "super-colour" given by its colouring pattern, one of c^m values. By the induction hypothesis on L, among the first N blocks there are blocks $B_0, B_1, \dots, B_{L-2}$ with indices $s, s+t, \dots, s+(L-2)t$ ($t \geq 1$) that are *identically coloured*. Inside B_0 apply Claim($r-1$) to the relative positions $1, \dots, m$: either there is a monochromatic L-AP (done), or there are $r-1$ colour-focused $(L-1)$-APs with relative starts a_i, steps d_i, and common relative focus $f = a_i + (L-1)d_i \leq m$. Since no monochromatic L-AP arose, f's colour differs from every A_i's colour.
  Define, in absolute positions, for $i < r$: $A'_i = \{sm + a_i + k(d_i + tm) : 0 \leq k \leq L-2\}$. Its k-th element is the relative position $a_i + kd_i$ inside block B_k (block B_k starts at $(s+kt)m$), and B_k is coloured like B_0, so A'_i is monochromatic in the colour of A_i. Define $A'_r = \{sm + f + ktm : 0 \leq k \leq L-2\}$, the position f in each B_k: it is monochromatic in the colour of f. The r colours are distinct, and each A'_i has focus $sm + f + (L-1)tm$ (for $i < r$ because $a_i + (L-1)d_i = f$). That focus is at most $(s + (L-1)t + 1)m \leq (2N+1)m =: M(r)$. So Claim($r$) holds. ∎

3 Aside on the same board: the IMO 1988 Vieta-jumping problem

The box at the very top of the board is an unrelated problem. I read it as the numerator and denominator written with the roles swapped, but it is surely this one: *if a, b are positive integers*

and $k = \dfrac{a^2+b^2}{ab+1}$ *is an integer, then* k *is a perfect square* (IMO 1988, Problem 6).

Proof (Vieta jumping). Suppose k is not a square. Among all positive-integer solutions of $a^2 - kab + b^2 = k$ take one with $a+b$ minimal, and $a \geq b$. As a quadratic in a it has roots a and $a' = kb - a = \dfrac{b^2-k}{a}$. The number a' is an integer. If $a' = 0$ then $b^2 = k$, a square. If $a' \leq -1$ then $a'^2 - ka'b + b^2 \geq 1 + k + 1 > k$, contradicting the equation. So $a' \geq 1$, and $a' = \frac{b^2-k}{a} < \frac{b^2}{a} \leq b \leq a$. Then (a', b) is a solution with a smaller sum. Contradiction. □

Sources

- Van der Waerden's theorem (Wikipedia): the value $W(2,3) = 9$ and the double-induction outline.
- Proof of van der Waerden's theorem (opentext.uleth.ca) and Blondal–Junić, *Proof of van der Waerden's Theorem in Nine Figures*: the colour-focusing proof.
- The $W(2,3)$ case analysis above is my own and was checked by brute force: the extremal colouring of $1, \ldots, 8$ is R R B B R R B B (or its colour swap).